

Miguel G. SILVA

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SUMMARY

PhD researcher and applied AI scientist. I work where research and engineering meet: framing the hard modeling problem (e.g., neural surrogates for physics simulation, multi-objective constrained optimization over engineered products), then carrying the answer into production. Strong background in time-series modeling, pattern discovery, and interpretable ML, with first-author publications in Q1 journals.

WORK EXPERIENCE

2025 – PRESENT	Applied AI Scientist at DAREDATA ENGINEERING - Co-led the design of a production agentic engineering-to-order loop for a global automotive cable manufacturer, replacing a 16-week / ~€10k physical prototype-and-lab cycle with a 1-day / ~€1.8k software workflow (~80× faster, 5–10× cheaper per RFP). - Formulated and built the surrogate-modeling layer: a Graph Neural Network with calibrated uncertainty, trained on multiphysics simulation history and turning hours of physics solves into millisecond inference. - Designed the multi-objective constrained optimization over a mixed continuous/discrete design space (material × geometry × stack), balancing cost and performance under regulatory and manufacturability constraints. Calls the surrogate as a fast oracle, routing survivors to FEM validation. - Partnered with the client on product direction: framed engineering requirements as research problems, defined the tools and skills exposed by the engineer assistant, and informed roadmap trade-offs across the loop.
2021 – 2026	Doctoral Researcher at LASIGE & INESC-ID - Analyzed smart meter data from 2,170 households using biclustering, uncovering statistically significant and actionable local water consumption patterns inaccessible to traditional clustering methods. - Developed a scalable methodology to extract population density patterns from mobile phone data (3,743 cells) across Lisbon, uncovering multi-level seasonality and emerging urban trends. - Designed a methodology to extract recurring, non-trivial, and statistically significant patterns in multivariate time series data. - Published 3 first-author papers in top-quartile (Q1) scientific journals.
2022 – 2024	Teaching Assistant at IST, UNIVERSITY OF LISBON - Selected as invited TA to deliver 140+ hours of lectures and labs on Information Retrieval (Python) and Database Systems (SQL) to over 300 students. - Maintained an average student rating of 4.5/5 across three semesters.

EDUCATION

2021 - EXPECTED 2026	PhD in Computer Science <i>Faculty of Sciences University of Lisbon, Portugal</i>
2018 - 2020	MSc in Data Science <i>Faculty of Sciences, University of Lisbon, Portugal</i> GPA: 18 / 20 , THESIS: 19/20

TECHNICAL SKILLS

Languages & Core: Python · SQL · Git · LaTeX

ML / Deep Learning: PyTorch · Scikit-learn · Optuna · Pandas · NumPy

Applied AI: Surrogate Modeling · Multi-Objective Optimization · Time-Series Modeling · Uncertainty Quantification · Agentic AI

Engineering & MLOps: FastAPI · Pydantic · Docker · CI/CD (GitHub Actions) · MLflow · Async Pipelines · ETL

PROJECTS

2026	Statistical Significance for Multivariate Time Series Motifs Open-sourced <code>msig</code>, a Python package implementing statistical significance criteria that separate genuine recurring structure from spurious patterns in multivariate time series motif analysis. PyPI GitHub
2025	Motif Mining in Multivariate Time Series Built an unsupervised pipeline to mine recurring patterns in residual time series, revealing actionable behavioral insights in energy use, human mobility, and urban activity. Publication GitHub
2025	Next-Motif: Predicting Irregular Pattern Recurrence in Time Series Leveraged LSTM, TCN, and Transformer models to build a forecasting system for irregular event patterns in time series, reducing prediction error by 21–24% on urban mobility and electricity consumption data. Publication GitHub
2024	UrbanFlow: Spatiotemporal Analysis of Population Density Data Developed a Python-based, Docker-deployable tool to extract and visualize spatiotemporal population trends from mobile network data (3,743 cells), enabling actionable insights for data-driven urban planning. Publication GitHub
2023	Biclustering Water Consumption Patterns from Smart Meter Data Applied biclustering to household water data from a luxury resort (2,170 users), uncovering high-resolution seasonal consumption profiles to support anomaly detection and data-driven resource planning. Publication